

INTRODUCTION

I received my **M.S. from Pohang University of Science and Technology (POSTECH)** in August 2025. Since September 2025, I have been serving in South Korea as **Professional Research Personnel** (Alternative Military Service) at **60 Hertz** with an expected completion in 2028.

I am highly motivated to analyze **financial phenomena through data-driven research**. Throughout my undergraduate and graduate curriculum, I have utilized a wide range of approaches, from classical statistical techniques to state-of-the-art AI. With these experiences, I'm eager to study financial phenomena from two complementary angles: **(i) an engineering viewpoint** focused on improving analytical performance and **(ii) an economic perspective** aimed at deriving practical, real-world implications. With this background and these research interests, I aim to make meaningful contributions to the field.

* This CV is updated on 5 May 2026.

RESEARCH INTERESTS

Financial Domain; Data Science; Statistical Modeling; Deep Learning.

EDUCATION

• POSTECH <i>M.S. in Dept. of Industrial and Management Engineering</i> - Completed the Industrial AI program Thesis Title: Systemic Cyber Risks and Insurance Regulatory Capital.	Pohang, South Korea <i>Sep. 2023 - Aug. 2025</i>
• Inha University <i>B.S. in Dept. of Industrial Engineering</i>	Incheon, South Korea <i>Mar. 2019 - Aug. 2023</i>

EXPERIENCE

• 60 Hertz <i>Data Scientist (as a Technical Research Personnel)</i> - Developing renewable power generation forecasting models, with applications in anomaly detection and electricity market bidding solutions. Skill used: Python, SQL, Dagster. Communication Tool: Slack / Jira, Confluence.	Seoul, South Korea <i>Jul. 2025 - <u>Current</u></i>
• AIRM Lab., POSTECH <i>Graduate Researcher (Advisor: Kwangmin Jung)</i> - Conducted scenario-based analyses on how systemic cyber risk affects insurance regulatory models. Teaching Assistant (TA): Financial Accounting (IMEN203), Spring 2024; Spring 2025. Skill used: Python, R. Communication Tool: Slack.	Pohang, South Korea <i>Sep. 2023 - Jul. 2025</i>
• Informatics and Deep Learning Lab., Inha Univ. <i>Research Intern (Advisor: Wookey Lee)</i> - Analyzed performance degradation in Deep Learning (e.g., Multi-modal and Diffusion generative models) under adversarial attacks on input distributions. Skill used: Python / Keras, TensorFlow.	Incheon, South Korea <i>Apr. 2021 - Jun. 2023</i>

• **SL Solution Co. Ltd.**

Seoul, South Korea

Internship course at Software Team

Aug. 2020

- Developed a **responsive web application** visualizing COVID-19 patient movement paths on interactive maps by integrating public datasets with **geocoding** through the **Tmap REST API**.
- **Skill used:** Java, Javascript / SpringBoot.

• **UDMTEK Co. Ltd.**

Suwon, South Korea

Internship course at MES Team

Dec. 2019 – Feb. 2020

- Managed database on **MES**(Manufacturing Execution System) and developed a **MVC**(Model, View, and Controller) service for maintenance.
- **Skill used:** Java, TypeScript / Angular, SpringBoot.

PUBLICATIONS

A. Working Papers

- [1] **Bae, K.**, Jung, K., and Zhang, L. (2025). Systemic cyber risks and insurance regulatory capital. Submitted.

B. International Conferences

- [1] **Bae, K.**, Jung, K., and Zhang, L. (2025, Aug.). Systemic Cyber Risks and Insurance Regulatory Capital. *In the 5th World Risk and Insurance Economics Congress (**WRIEC**)*.
- [2] **Bae, K.**, Lee, S., and Lee, W. (2023, Dec.). Diffusion-C: Unveiling the Generative Challenges of Diffusion Models through Corrupted Data. *In the 37th Annual Conference on Neural Information Processing Systems (**NeurIPS**), Workshop on Diffusion Models*.
- [3] Lee, W., **Bae, K.**, Toshpulatov, M., Lee, S. (2022, Jul.). Bimodal Classification with CNN and Sequence Models. *In the 18th Int. Conference on Data Science (**ICDATA**)*.
- [4] **Bae, K.**, Lee, S., and Lee, W. (2022, Jan.). Transformer Networks for Trajectory Classification. *In the 2022 IEEE International Conference on Big Data and Smart Computing (**BigComp**)*.

C. Domestic Journals

- [1] Jung, K. and **Bae, K.** (2025). A Classification and Statistical Analysis on Systemic Cyber Risks. *Korean Journal of Insurance (**KJI**)*, 142, p.1-34.

D. Domestic Conferences

- [1] **Bae, K.**, Jung, K., Zhang, L. (2025, Aug.). Systemic cyber risks and insurance regulatory capital. *In the Korean Insurance Academic Society (**KIAS**)*.
- [2] Jung, K. and **Bae, K.** (2025, Feb.). A Classification and Statistical Analysis on Systemic Cyber Risks. *In the 3rd Social Science Korea (**SSK**) Networking Symposium*.
- [3] Jung, K. and **Bae, K.** (2025, Feb.). A Classification and Statistical Analysis on Systemic Cyber Risks. *In the Korean Insurance Academic Society (**KIAS**)*.
- [4] **Bae, K.**, Lee, S., and Lee, W. (2022, Dec.). Robustness Analysis of Diffusion Generation model Using Noises and Corruptions. *In the Korean Software Congress (**KSC**)*.
- [5] **Bae, K.**, Lee, S., and Lee, W. (2022, Jul.). Robust Multimodal Classification Model Using Homogeneous Features. *In the Korean Computer Congress (**KCC**)*.

● **AI-driven Bidding Strategy Solution for the Japan Electricity Market.**

SK Innovation Co., Ltd. (SKI)

*Jan. 2026 - **Current***

- **Keywords:** JEPX, MLOps, Operations Research, Risk Management.
- Developing an AI-driven bidding strategy solution provided as an API for the Japan Electricity Power Exchange (JEPX), supporting PV, ESS, and hybrid PV-ESS operators.
- Built end-to-end pipelines using Dagster deployed on Kubernetes, (i) collecting weather data from Open-Meteo, supply-demand data from OCCTO, and market price data from JEPX, and (ii) storing forecasted price data from machine learning models.
- Implemented machine learning models to forecast day-ahead and intraday electricity prices based on meteorological forecasts, historical demand data, and past price data.
- Proposed a MILP-based operations research framework to offer a bidding solution that manages bid rejection risk in the JEPX day-ahead market.

● **Data-Driven Maximum Power Output Forecasting for Combined Cycle Power Plants.**

Korea Midland Power Co., Ltd. (KOMIPO)

Oct. 2025 - Apr. 2026

- **Keywords:** Thermal Power Plants, Maximum Power Output, Hybrid Machine Learning.
- Proposed a data-driven AI model to forecast unit-level maximum power output for thermal power plants.
- Predicted the maximum power output based on LDAPS meteorological forecasts (temperature, relative humidity) and operational data (elapsed days since turbine overhaul and water washing).
- Built a hybrid machine learning architecture, utilizing Linear Regression for baseline prediction and a LightGBM regressor to minimize residual errors.

● **Assessment of the Corporate Customer Service and Proposal for the Improvement.**

Korean Fire Protection Association (KFPA)

Sep. 2024 - Nov. 2024

- **Keywords:** Linear Programming, Resource Allocation, Operations Research.
- Proposed an optimal workforce allocation strategy across employment types (e.g., subsidiary, contract, and outsourcing) to enhance the operational efficiency of fire safety inspections conducted by KFPA.
- Applied an operations research framework, formulating a linear programming model that jointly accounts for productivity and labor costs across employment types.
- Identified contract workers as the most cost-effective employment type for fire safety inspections through the proposed model.

● **Data-driven Evaluation for Safety Assessment of the KFPA and its Future Strategy.**

Korean Fire Protection Association (KFPA)

Apr. 2024 - Jul. 2024

- **Keywords:** Random Forest, Explainable AI, SHAP.
- Quantified the impact of digital transformation and AI adoption (introduced in 2022) on the workload of fire safety inspectors at KFPA, complemented by on-site field observations of inspection processes.
- Developed a random forest model to predict inspection man-day (MD) consumption from building characteristics, and applied SHAP-based explainable AI techniques to quantify the contribution of each factor.
- Identified distinct inspection time drivers across building types and confirmed a significant workload reduction following the digital transformation.

● **Responsive Web Application Development.**

SL Solution

Aug. 2020

- **Keywords:** REST API, Geocoding, Responsive Web Service.
- Developed a responsive web application to store and visualize the location data of COVID-19 confirmed cases. [\[Github\]](#)
- Used the web development tools, including Spring Boot, MariaDB, and T-Map API.

● **MES Web Application Maintenance Project.**

UDMTEK

Dec. 2019 - Feb. 2020

- **Keywords:** MES Service, Responsive Web Service.
- Developed a web application for Manufacturing Execution System (MES) services.
- Implemented the MVC pattern using web development frameworks (e.g., Angular, Spring Boot, and PostgreSQL).

SKILLS

- **Language:** Korean (Native), English (Advanced).
- **Programming:** Python, R, C++ (Advanced), Java, JavaScript, SQL (Intermediate).
- **AI Framework:** TensorFlow, PyTorch, Scikit-Learn (Advanced).